

* Measure of central tendency

Measure of central tendency are statistical metrics that describe the centre point or typical value of a dataset. They provide a single value that summarizes a set of data by identifying the central position within the dataset.

1. Mean / Average

2. Median

3. Mode

① Mean:-

Sum of values divided by the no. of values

Population mean (μ)

Population (N)

$$\mu = \frac{\sum_{i=1}^N X_i}{N}$$

Sample mean ($\bar{x}$)

Sample (n)

$$\bar{x} = \frac{\sum_{i=1}^n X_i}{n}$$

$X \rightarrow$ random variable

$$X = \{5, 8, 12, 15, 20\}$$

$$N = 5$$

$$\mu = \frac{5 + 8 + 12 + 15 + 20}{5}$$

Characteristics:-

i. Mean is usually affected by extreme outliers.

ii. Used for interval & ratio data (Refer intro chapter)

$$X = \{1, 2, 3, 4, 5\}$$

$$\mu = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

$$X = \{1, 2, 3, 4, 50\}$$

$$\mu = \frac{1 + 2 + 3 + 4 + 50}{5} = 108$$

② Median:-

Middle value in a dataset when values are arranged in asc/desc order

$$X = \{1, 2, 3, 4, 5\}$$

↑

$$X = \{5, 2, 1, 100, 3, 4\}$$

$$\Rightarrow \{1, 2, 3, 4, 5, 100\}$$

$$= \frac{3+4}{2} = 3.5$$

Characteristics:-

i. not affected by extreme outliers

ii. Used specifically for ordinal, interval & ratio data (refer intro chapter)

③ Mode:-

Value that appears most frequently in a dataset

$$\underline{2}, \underline{4}, \underline{4}, 6, 8, 9, 10, \underline{2}, 3$$

$$\text{mode} = 2, 4 \text{ (bimodal)}$$

Characteristics:-

i. not affected by outliers

ii. Used for Nominal, ordinal, interval and ratio data

Real world applications:-

Feature engineering

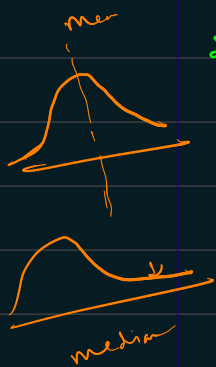
Age	Weight	Salary	Gender	Degree
24	55	2	M	BSc
28	60	3	F	—
32	80	5	F	—
40	—	7	F	Phd
—	—	1	—	BSc
—	90	2	M	MS
55	75	—	—	—

i) Handling the missing values

→ If the plot of age is symmetrical, we can replace the missing value by mean.
→ If right/left skewed, then by median.

mode for
nominal

mode for
nominal &
ordinal



§ Measure of Dispersion:-

describe the spread or variability of a dataset. They indicate how much the values in a dataset differ from the central tendency

Common measures of dispersion:-

1. Range
2. Variance
3. Standard Deviation
4. Interquartile Range (IQR)

1. Range:-

Difference b/w the max & min values in a dataset.

$$\text{Range} = \text{max value} - \text{min value}$$

Characteristics:-

- i. Simple to calculate
- ii. Sensitive to outliers
- iii. Rough measure of dispersion (Doesn't say about the middle values)

2. Variance:-

Measures the average squared deviation of each value from mean.
provides a sense of how much a value in a dataset vary

Population variance

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

Sample variance

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

Ex: {5, 8, 12, 15, 20}

$$N = 5$$

$$\mu = \frac{5 + 8 + 12 + 15 + 20}{5} = 12$$

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

$$\sigma^2 = \frac{(5-12)^2 + (8-12)^2 + (12-12)^2 + (15-12)^2 + (20-12)^2}{12}$$

$$= \frac{49 + 16 + 9 + 64}{12} = 27.6$$

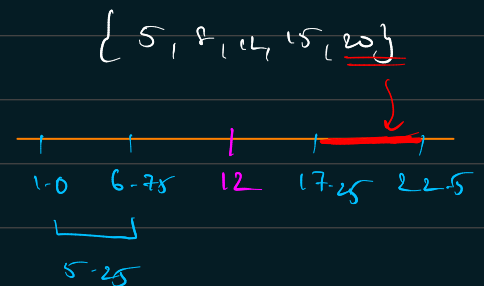
Characteristics :-

- i. Provides a precise measure of variability -
- ii. Units are squared of the original data units -
- iii. more sensitive to outliers compared to range (b/c we are squaring)

3- Standard deviation :-

Square root of variance -

$$\sigma = \sqrt{27.6} \approx 5.25$$



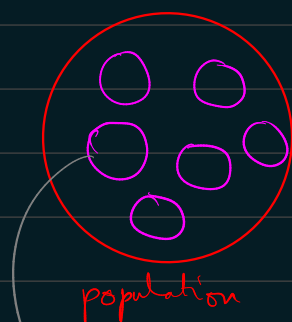
Characteristics :-

- i. Provides a clear measure of spread in the same units as the data.
- ii. Sensitive to outliers
- iii.

Sample variance :-

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1} \rightarrow ?$$

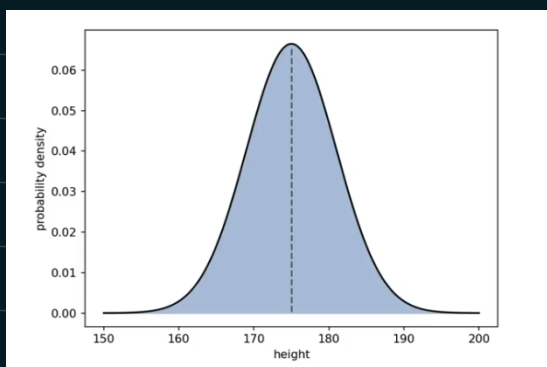
$\Rightarrow$ Bessel's correction



$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

$$s^2 = \sum_{i=1}^n \frac{(x_i - \mu)^2}{n}$$



$$\mu = 175$$

$$\sigma^2 = 36$$

Sample 1 :- [173, 173, 172, 161]

$$\bar{x} = 170.38$$

$$s^2 = \frac{\sum_{i=1}^n (\underline{x_i} - \bar{x})^2}{\underline{n}}$$

$$s^2 = 23.84$$

Sample 2 :- [164, 175, 177, 188]

$$\bar{x} = 171.29$$

$$s^2 = 28.07$$

$$\frac{s_1^2 + s_2^2 + \dots + s_{1000}^2}{1000} \approx 27 < \sigma^2$$

$$27 \cdot \frac{4 \rightarrow n}{3 \rightarrow (n-1)} = 36 = \sigma^2$$

→ (n-1) is divided instead of 'n' so as to not underestimate the population variance.

★ Random variables :-

X → function whose value is derived from different process.

Tossing a coin :

$$X = \begin{cases} H \\ T \end{cases}$$

Rolling a dice

$$x = \begin{cases} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{cases}$$

Random variable

Discrete random variable

Ex :- Tossing a coin
Rolling a dice

Continuous random variable

Amount of rainfall next year
Height of people

* Percentile and Quartile

Percentage :-

$\{1, 2, 3, 4, 5\}$

odd no.s = 3

$$\% \text{ of odd no.s} = \frac{3}{5} \times 100 = 60\%$$

Percentile :-

Percentile is a value below which certain percentage of observations lie.

$\{2, 2, 3, 4, 5, 6, 7, 7, 8, 9\}$

$$\text{Percentile of value } x = \frac{\text{\# of values below } x}{n} \times 100$$

$$\text{" } 7 = \frac{6}{10} \times 100$$

$$= 60$$

↓

60% of all values are below are 7.

→ To find the value from percentile

$$\text{Value} = \frac{\text{Percentile}}{100} \cdot (n+1)$$

$$\Rightarrow \frac{25}{100} (11) = 2.75$$

Quartile :-

25% = 1st Quartile

50% = 2nd Quartile

75% = 3rd Quartile

* 5 number Summary

1) minimum

2) 1st Quartile (25%)

3) Median

4) 3rd Quartile (75%)

5) Maximum

Q: 1, 1, 1, 2, 2, 2, 2, 3, 3, 4, 5, 5, 5, 5, 6, 7, 7, 9, 32

Lower fence $\longleftrightarrow$ Higher fence

$$\text{Lower fence} = Q_1 - 1.5(IQR)$$

$$\text{Higher fence} = Q_3 + 1.5(IQR)$$

IQR $\rightarrow$ Inter Quartile Range ($Q_3 - Q_1$)

$$\begin{aligned} Q_1 &= \frac{\text{Percentile}}{100} \times (n+1) \\ &= \frac{25}{100} \times (20) = 5^{\text{th}} \text{ position} = 2 \end{aligned}$$

$$Q_3 = \frac{75}{100} \times (20) = 15^{\text{th}} \text{ position} = 6$$

$$IQR = Q_3 - Q_1 = 6 - 2 = 4$$

$$\begin{aligned} LF &= Q_1 - 1.5(IQR) \\ &= 2 - 1.5(4) \\ &= -4 \end{aligned}$$

$$\begin{aligned} HF &= Q_3 + 1.5(IQR) \\ &= 6 + 1.5(4) \\ &= 12 \end{aligned}$$

$\Rightarrow$ Any value higher than HF or lower than LF is considered outlier

Ex: 1, 1, 1, 2, 2, 2, 2, 3, 3, 4, 5, 5, 5, 5, 6, 7, 7, 9, 32 → outlier (not considered)

Minimum = 1

$Q_1 = 2$

Median = 5

$Q_3 = 6$

Maximum = 9

✱ Histogram

a graphical representation of the distribution of numerical data. It is an estimate of the probability distribution of a continuous variable and is used to visualize the shape, central tendency and variability of a dataset.

Ex: Age = { 6, 7, 8, 8, 9, 12, 14, 15, 15, 17, 17, 17, 28, 35, 50 }
(0 - 50)

no. of bins = 10 $\Rightarrow \frac{50}{10} = 5 \rightarrow$ bin size

Bins $\rightarrow$ [0-5, 6-10, 11-15, 16-20, 21-25, 26-30, ..., 46-50]

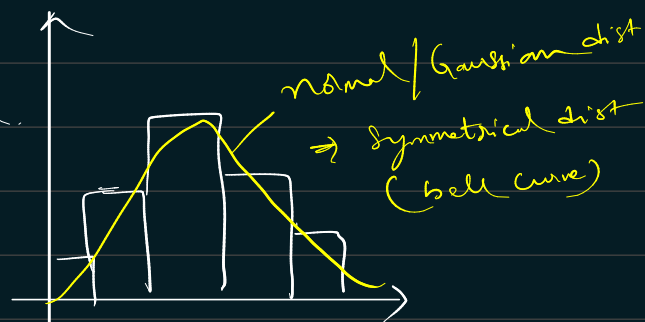


✱ Skewness :-

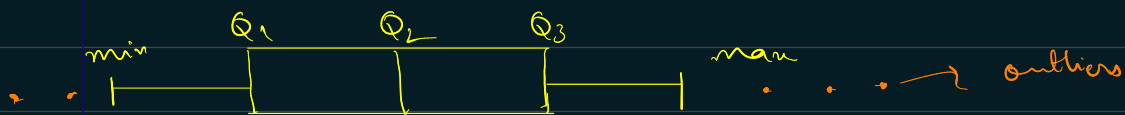
→ No skewness in bell curve / symmetrical.

→ 50% to left

50% to right



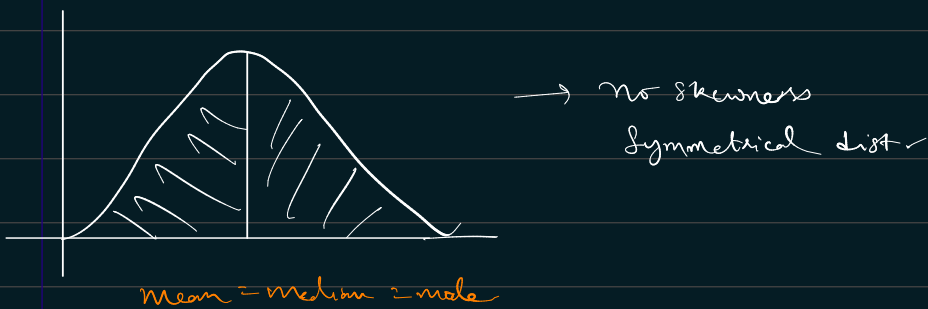
Box plot



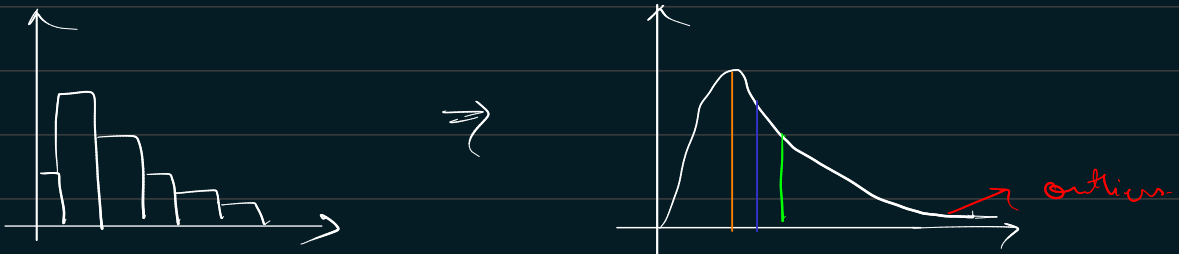
⇒ For a symmetrical dist., $Q_3 - Q_2 \approx Q_2 - Q_1$

⇒ The mean, median & mode are all perfectly at the centre.

① No skewness



② Right skewed / positive skewed



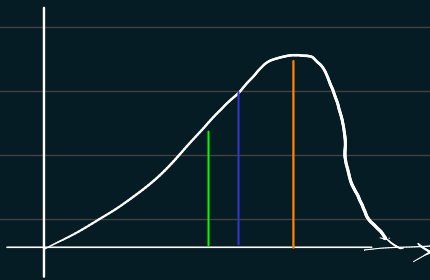
Ex:- Log Normal distribution

mean $>$ mode $>$ mode

Box plot:-

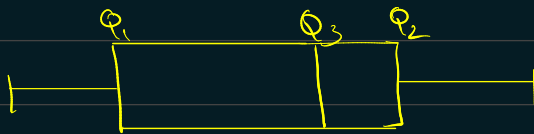


② Left skewed / negative skew



$$\text{mean} < \text{median} < \text{mode}$$

box plot :-



$$Q_2 - Q_1 > Q_3 - Q_2$$

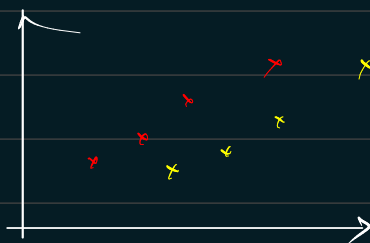
* Covariance and Correlation:-

Covariance and correlation are two statistical measures used to determine the relationship between two variables. Both are used to understand how changes in one variable are associated with changes in another variable.

Covariance :-

Covariance is a measure of how much two random variables change together. If the variables tend to increase and decrease together, the covariance is positive. If one tends to increase when the other decreases, the covariance is negative.

X	Y	
2	6	X ↑ Y ↑ +
3	7	X ↓ Y ↑ -
5	9	X ↑ Y ↓ -
7	12	X ↓ Y ↓ +



$x \uparrow \quad y \uparrow$
 $x \downarrow \quad y \downarrow$

$\Rightarrow$ +ve Covariance

$$\text{Cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\Rightarrow \text{Cov}(x, x) = \sum_{i=1}^n \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$$

$$\boxed{\text{Cov}(x, x) = \text{Var}(x)}$$

Ex:

Hours Studied

Exam Score

2

50

3

60

4

70

5

80

6

90

$\Rightarrow \begin{matrix} x \uparrow & y \uparrow \\ x \downarrow & y \downarrow \end{matrix} \Rightarrow \text{true Covariance}$

$$\text{Cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\bar{x} = 4$$

$$\bar{y} = 70$$

$$\text{Cov}(x, y) = \frac{(-2)(-20) + (-1)(-10) + (0)(0) + (1)(10) + (2)(20)}{4}$$

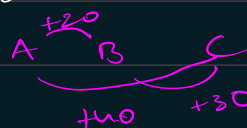
$\text{Cov}(x, y) = 25 \Rightarrow$ true Cov indicates no. of hrs studied increases the score in exam

Advantages

1. Quantify the relationship b/w x & y

Disadvantages

i. doesn't have a specific limit value



$\text{Cov}(A, B) = 20$ $\text{Cov}(B, C) = 30$
 $\Rightarrow$ Can't be compared

$\text{Cov}(x, y) \Rightarrow -\infty \text{ to } \infty$

* Correlation $\begin{cases} \rightarrow \text{Pearson Correlation Coefficient} \\ \rightarrow \text{Spearman Rank Coefficient} \end{cases}$

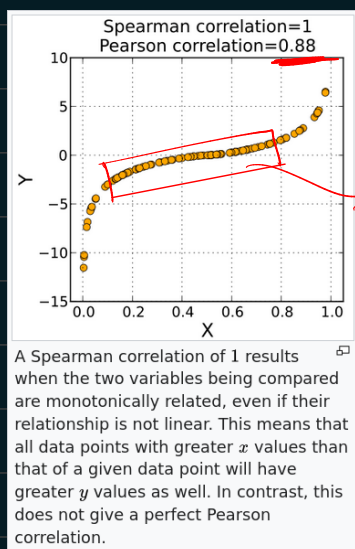
① Pearson Correlation Coefficient $\Rightarrow [-1 \text{ to } 1]$

$$r_{x,y} = \frac{\text{Cov}(x,y)}{\sigma_x \sigma_y}$$

$\rightarrow$ The more the value towards +1, the more +ve correlated x & y

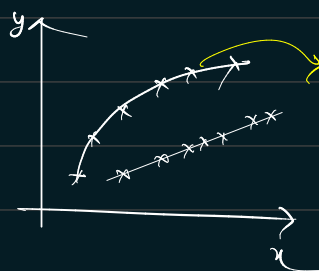
$\rightarrow$ The more the value towards -1, the more -ve correlated x & y

② Spearman Rank Correlation:



$\Rightarrow x \uparrow y \uparrow$
 $x \downarrow y \downarrow$

The rate of Δx to Δy is smaller compared to other points.



Pearson correlation is not able to capture the correlation precisely for non linear data.

$$r_s = \frac{\text{Cov}(R(x), R(y))}{\sigma(R(x)) \cdot \sigma(R(y))}$$

